MATH 6660 — Homework 4

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Problem 1

Download the PyDistMesh mesh generator. You can find all the necessary links from the Math 6660 Canvas page.

PyDistMesh, B. Froehle's Python port of P.-O. Persson's DistMesh, was downloaded from the Canvas page and is used unmodified in Problems 2 and 3. Table 1 lists the files in the package.

file	role
`distmesh2d.py`	`p,t = distmesh2d(fd,fh,h0,bbox,pfix)`: level set `fd`, relative size `fh`, edge length `h0`; returns nodes `p` ($N \times 2$) and triangles `t` ($M \times 3$)
`distance_functions.py`	level sets (`dcircle`, `drectangle`, ...) and Booleans: `dunion` = min, `dintersect` = max, `ddiff`$(d_1, d_2) = \max(d_1, -d_2)$
`boundary.py`	`bnd2d`: re-triangulates, marks nodes with $\lvert\phi\rvert \leq 10^{-8}$ as boundary and puts them last, returns the areas
`assemble.py`	the CG(1) code used in Problem 3
`example1.py`, `example2.py`, `poisson_example.py`	unit disc; square with a hole, graded; $-\Delta u = 1$ on the latter

Table 1: The PyDistMesh bundle as downloaded (plus two helper files, `utils.py` and `mlcompat.py`).

Problem 2

Use this code to generate a mesh for a star-shaped region whose boundary is the zero contour of $\phi(r,\theta) = r - 0.75 - 0.25\sin(5\theta)$.

The level set in Cartesian coordinates

PyDistMesh evaluates the level-set function `fd` at Cartesian points, so ϕ is rewritten with $r = \sqrt{x^2+y^2}$ and $\theta = \tan^{-1}(y/x)$, taken in the quadrant of the point (x,y):

$$\begin{gathered}\phi(x,y) = \sqrt{x^2+y^2} - 0.75 - 0.25\sin\bigl(5\tan^{-1}(y/x)\bigr),\\ \phi < 0 \iff r < R(\theta) := 0.75 + 0.25\sin 5\theta.\end{gathered} \tag{2.1}$$

The boundary radius $R(\theta)$ lies between 0.5 and 1, so the star has five tips at $r = 1$ and five valleys at $r = 0.5$, and the box $(-1,1)^2$ contains Ω. The mesh is generated by

```
fd = lambda p: np.sqrt(p[:,0]**2 + p[:,1]**2) - 0.75 \
               - 0.25*np.sin(5*np.arctan2(p[:,1], p[:,0]))
p, t = dm.distmesh2d(fd, distfunc.huniform, h0, (-1,-1,1,1))
p, t, nin, nbd, area = bnd.bnd2d(fd, p, t, -1.0, -1.0, 1.0, 1.0)
```

To check the mesh, the exact area of the star is computed in polar coordinates:

$$|\Omega| = \int_0^{2\pi}\int_0^{R(\theta)} r\,\mathrm{d}r\,\mathrm{d}\theta = \frac{1}{2}\int_0^{2\pi}\Bigl[0.75^2 + 1.5\cdot 0.25\underbrace{\sin 5\theta}_{\int=0} + 0.25^2\underbrace{\sin^2 5\theta}_{\int=\pi}\Bigr]\mathrm{d}\theta = \frac{19\pi}{32} = 1.86532. \tag{2.2}$$

The boundary nodes

`bnd2d` marks a node as boundary only if $|\phi| \le 10^{-8}$, and Problem 3 imposes $u = 0$ only at those nodes. On the star, 66 of the 138 nodes on boundary edges sit just inside the curve, $\phi \approx -10^{-4}$, around the valleys (red in Figure 1), and are missed. Before `bnd2d`, every node on a boundary edge is therefore moved radially onto the curve,

$$(x,y) \;\leftarrow\; R(\theta)\,(\cos\theta,\sin\theta), \qquad \text{so} \quad \phi = 0, \tag{2.3}$$

a move of at most $10^{-4} \ll h_0$. Afterwards every boundary node is marked (Table 2).

The mesh

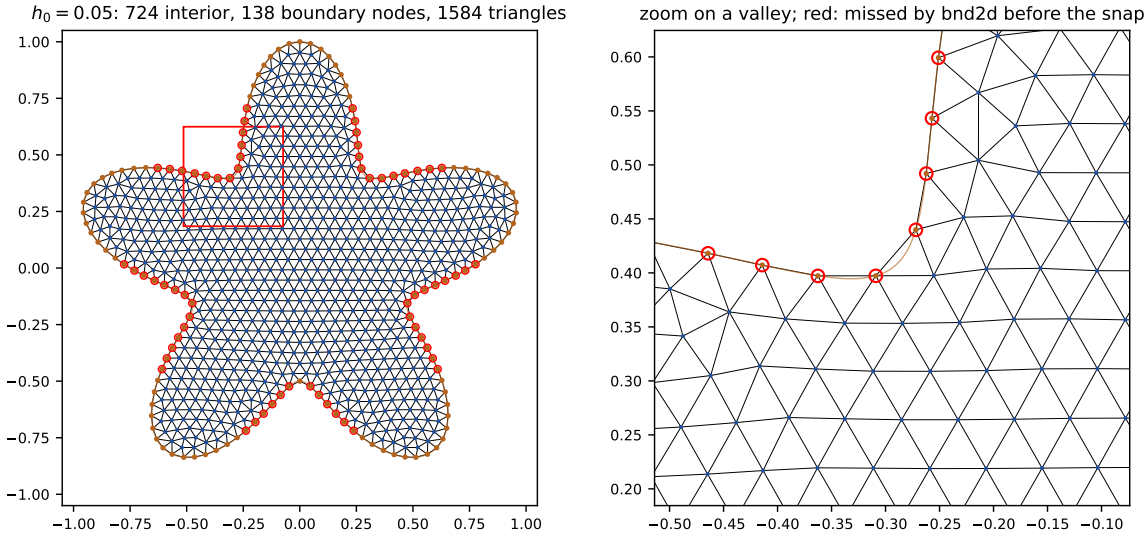


Figure 1: The star mesh at $h_0 = 0.05$, after the fix (2.3). Blue: interior nodes; brown: boundary nodes; the curve is $\phi = 0$. Red circles: the boundary nodes that `bnd2d` classed as interior before the fix.

h_0	interior	boundary	elements	mesh area	rel. error	$\max_j h_j/\rho_j$	flagged: raw $\to$ snapped
0.100	149	66	362	1.86001	-2.8×10^{-3}	2.57	$32/66 \to 66/66$
0.050	724	138	1584	1.86432	-5.4×10^{-4}	2.46	$72/138 \to 138/138$
0.025	3167	282	6614	1.86498	-1.9×10^{-4}	2.45	$140/282 \to 282/282$

Table 2: Star meshes. Exact area $19\pi/32 = 1.86532$ (2.2). Last column: nodes `bnd2d` marks as boundary, out of the nodes on boundary edges, before and after (2.3).

Table 2 shows the area error falling from 0.3% to 0.02% as h_0 decreases from 0.1 to 0.025. The ratio $\max_j h_j/\rho_j$ stays at or below 2.57 on every mesh, against 1.73 for an equilateral triangle, so the meshes are regular in the sense of (1.260) and have no slivers in the valleys.

Problem 3

Use the CG(1) method developed in class and provided on the Math 6660 Canvas page to solve $-u_{,xx} - u_{,yy} = 1$ *in* Ω, $u = 0$ *on* $\partial\Omega$, *on the star of Problem 2.*

The method

Let $V = H_0^1(\Omega)$. The weak form of the problem is: find $u \in V$ such that

$$\iint_\Omega \nabla u \cdot \nabla v \,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v \,\mathrm{d}x\,\mathrm{d}y \qquad \text{for all } v \in V, \tag{3.1}$$

with $f = 1$. Let V_h be the space of continuous functions that are linear on each triangle of the Problem 2 mesh and vanish at the boundary nodes. Such functions have square-integrable gradients and vanish on the boundary, so $V_h \subset V$. (Strictly, the mesh covers a polygon that differs from Ω by slivers of width $\mathcal{O}(h^2)$ along the curved boundary; this difference is neglected.) The CG(1) approximation U is obtained by restricting (3.1) to this subspace: find $U \in V_h$ such that

$$\iint_\Omega \nabla U \cdot \nabla v \,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v \,\mathrm{d}x\,\mathrm{d}y \qquad \text{for all } v \in V_h. \tag{3.2}$$

A basis of V_h is formed by the hat functions $\varphi_1, \ldots, \varphi_M$ of the M interior nodes. Writing $U = \sum_{i=1}^M \xi_i \varphi_i$ and taking $v = \varphi_j$ in (3.2), $j = 1, \ldots, M$, gives

$$\sum_{i=1}^{M} \underbrace{\iint_\Omega \nabla\varphi_i \cdot \nabla\varphi_j \,\mathrm{d}x\,\mathrm{d}y}_{A_{ji}} \xi_i = \underbrace{\iint_\Omega f\varphi_j \,\mathrm{d}x\,\mathrm{d}y}_{b_j}, \qquad \text{that is,} \quad \boldsymbol{A\xi} = \boldsymbol{b}, \tag{3.3}$$

where ξ_i is the value of U at interior node i.

What `assemble.py` computes on each triangle

On a triangle Ω_e with vertices (x_l, y_l), $l = 1, 2, 3$ and $|\Omega_e|$ is the area of this triangle, let φ_k be the hat function of vertex k ($k = 1, 2, 3$): it equals 1 at vertex k and 0 at the other two, so its value at vertex l is δ_{kl}. On Ω_e it has the form $\varphi_k = \varphi_k(x, y) = a_k + b_k x_l + c_k y_l = \delta_{kl}$. Evaluating it at the three vertices gives three equations for its coefficients:

$$\begin{aligned} a_k + b_k x_1 + c_k y_1 &= \delta_{k1}, \\ a_k + b_k x_2 + c_k y_2 &= \delta_{k2}, \\ a_k + b_k x_3 + c_k y_3 &= \delta_{k3}, \end{aligned} \qquad \text{that is,} \qquad \underbrace{\begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{bmatrix}}_{\boldsymbol{X}} \begin{bmatrix} a_k \\ b_k \\ c_k \end{bmatrix} = \begin{bmatrix} \delta_{k1} \\ \delta_{k2} \\ \delta_{k3} \end{bmatrix}. \tag{3.4}$$

Multiplying by $\boldsymbol{X}^{-1}$ therefore gives $(a_k, b_k, c_k)^T$ and $\nabla\varphi_k = \partial\varphi_k/\partial x, \partial\varphi_k/\partial y = (b_k, c_k)$ on Ω_e. Since the gradients are constant on Ω_e, the stiffness integral is computed exactly:

$$\begin{aligned} \iint_{\Omega_e} \nabla\varphi_j \cdot \nabla\varphi_k \,\mathrm{d}x\,\mathrm{d}y &= \iint_{\Omega_e} (b_j, c_j) \cdot (b_k, c_k) \,\mathrm{d}x\,\mathrm{d}y && \text{gradients on } \Omega_e \\ &= \iint_{\Omega_e} (b_j b_k + c_j c_k) \,\mathrm{d}x\,\mathrm{d}y && \text{dot product} \\ &= (b_j b_k + c_j c_k) \iint_{\Omega_e} \mathrm{d}x\,\mathrm{d}y && \text{a constant leaves the integral} \\ &= |\Omega_e| \,(b_j b_k + c_j c_k) && \textstyle\iint_{\Omega_e} \mathrm{d}x\,\mathrm{d}y = \text{area of } \Omega_e. \end{aligned} \tag{3.5}$$

The load vector uses the vertex rule, using trapezoidal rule applied on a triangle:

$$\iint_{\Omega_e} f\varphi_k \,\mathrm{d}x\,\mathrm{d}y \approx \frac{|\Omega_e|}{3}\sum_{l=1}^{3} f(x_l,y_l)\,\underbrace{\varphi_k(x_l,y_l)}_{\delta_{kl}} = \frac{|\Omega_e|}{3}\,f(x_k,y_k). \tag{3.6}$$

For $f \equiv 1$ this rule is exact: φ_k is linear, so its mean over Ω_e equals its value at the centroid, which is the average of its vertex values, $(1+0+0)/3$.
Only the nodes with index below `num_int_nodes` receive a row and a column of $\boldsymbol{A}$, so the condition $u = 0$ is built into V_h. The assembled matrix is exactly symmetric, $\max|\boldsymbol{A} - \boldsymbol{A}^T| = 0$.

The solution on the star

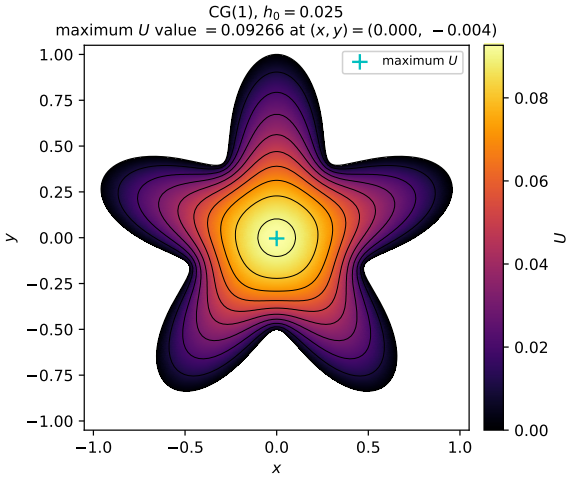


Figure 2: CG(1) solution of $-\Delta u = 1$, $u = 0$ on the star, $h_0 = 0.025$ (3167 unknowns), with ten contours. Cross: the largest nodal value.

h_0	unknowns	$\max U$	at (x, y)	$\max U$, raw mesh
0.100	149	0.09233	(0.000, 0.032)	0.23841
0.050	724	0.09256	(−0.026, −0.008)	0.22573
0.025	3167	0.09266	(0.000, −0.004)	0.22951

Table 3: The star under refinement. "Raw mesh": the same nodes without the fix (2.3) of Problem 2. None of these refinement ensure that center (0,0) get a node. It might be possible.

The largest nodal value is $\max U = 0.0927$. It occurs at the node nearest the centre, where the five-fold symmetry places it, and it is stable to three digits from $h_0 = 0.05$ on. Without the correction of Problem 2 the code returns $\max U \approx 0.23$. The nodes missed in Problem 2 are then left free, so along the valleys the code imposes no condition instead of $u = 0$, and the solution is wrong.

Note: on the raw mesh, `bnd2d` misses about half the boundary nodes, all of them around the five valleys. The solver treats those nodes as unknowns, so nothing holds $u = 0$ there. What the missed nodes get instead: a boundary node with no imposed value falls under the weak form's natural condition, zero normal flux, $\partial u/\partial n = 0$. In membrane terms, the rim is clamped only near the five tips and free along the valleys.

Problem 4(a)

Recast this problem as a variational problem. Clearly state the test and trial function spaces.

The strong problem is

$$-\nabla\cdot\nabla u = -u_{,xx} - u_{,yy} = f(x,y) \quad \text{in } \Omega = (-1,1)\times(-1,1), \qquad u = g \quad \text{on } \partial\Omega. \tag{4a.1}$$

Multiplying by a test function and integrating by parts [product rule, then divergence theorem]

Multiply (4a.1) by a test function v and integrate over Ω:

$$-\iint_\Omega v\,\Delta u\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v\,\mathrm{d}x\,\mathrm{d}y. \tag{4a.2}$$

By the product rule, $\nabla\cdot(v\nabla u) = \nabla v\cdot\nabla u + v\,\Delta u$, so the left-hand side of (4a.2) becomes

$$-\iint_\Omega v\,\Delta u\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega \nabla u\cdot\nabla v\,\mathrm{d}x\,\mathrm{d}y - \iint_\Omega \nabla\cdot(v\nabla u)\,\mathrm{d}x\,\mathrm{d}y. \tag{4a.3}$$

The divergence theorem, with $\boldsymbol{n}$ the outward unit normal, turns the last term into a boundary integral, $\iint_\Omega \nabla\cdot(v\nabla u)\,\mathrm{d}x\,\mathrm{d}y = \oint_{\partial\Omega} v\,\nabla u\cdot\boldsymbol{n}\,\mathrm{d}s$. Substituting this into (4a.3), and the result into (4a.2), gives

$$\iint_\Omega \nabla u\cdot\nabla v\,\mathrm{d}x\,\mathrm{d}y \;-\; \oint_{\partial\Omega} v\,\underbrace{\nabla u\cdot\boldsymbol{n}}_{\text{normal flux, not given}}\,\mathrm{d}s \;=\; \iint_\Omega f v\,\mathrm{d}x\,\mathrm{d}y. \tag{4a.4}$$

Removing the boundary term [Lecture 5: Dirichlet = essential, the test function vanishes there]

On all four sides the boundary data prescribe the *value* $u = g$. The normal flux $\nabla u\cdot\boldsymbol{n}$ is therefore not known there; it is the reaction needed to hold u at g. Consequently the boundary integral in (4a.4) cannot be evaluated, and the only way to remove it is to require $v = 0$ on $\partial\Omega$. With that requirement, (4a.4) reduces to

$$\boxed{\iint_\Omega \nabla u\cdot\nabla v\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v\,\mathrm{d}x\,\mathrm{d}y} \tag{4a.5}$$

The trial set and the test space [from lecture : $u(0)=\alpha\neq 0$ gives trial $V \neq$ test V_0;]

Equation (4a.5) requires $\iint_\Omega|\nabla u|^2$ and $\iint_\Omega|\nabla v|^2$ to be finite, so, from lecture, both spaces are built on functions with $\iint_\Omega\left[v^2+|\nabla v|^2\right]\mathrm{d}x\,\mathrm{d}y<\infty$. In addition, the unknown u must satisfy the essential condition $u = g$, while the test function must vanish on the boundary. This gives

$$\boxed{\begin{aligned} \text{trial:}\quad & V_g = \left\{v:\ \iint_\Omega\left[v^2+|\nabla v|^2\right]\mathrm{d}x\,\mathrm{d}y<\infty,\quad v=g \text{ on } \partial\Omega\right\},\\ \text{test:}\quad & V_0 = \left\{v:\ \iint_\Omega\left[v^2+|\nabla v|^2\right]\mathrm{d}x\,\mathrm{d}y<\infty,\quad v=0 \text{ on } \partial\Omega\right\} = H_0^1(\Omega), \end{aligned}} \tag{4a.6}$$

where $|\nabla v|^2 = (v_{,x})^2 + (v_{,y})^2$. The two sets differ because $g\neq 0$ (in part (4.d), $g = 1$). This is a careful scenerio: V_g is not a vector space, since the sum u_1+u_2 of two of its members has boundary value $2g$, whereas V_0 is a vector space.

(Variational problem) Find $u \in V_g$ such that

$$\iint_\Omega \nabla u \cdot \nabla v \,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v \,\mathrm{d}x\,\mathrm{d}y \qquad \text{for all } v \in V_0,$$

with trial set V_g and test space $V_0 = H_0^1(\Omega)$ from (4a.6).

Problem 4(b)

Show that the variational problem has a unique solution by showing that it meets all of the criteria of the Lax–Milgram theorem.

Reduction to the form of the Lax–Milgram theorem [Lecture 13: u and v in the same Hilbert space; Lecture 5: lift the boundary data]

The Lax–Milgram theorem of Lecture 13 requires a single Hilbert space V that contains both u and v. In 4(a), however, $u \in V_g$ while $v \in V_0$. Lecture 5 resolves this by splitting off a known function that carries the boundary data.

Let G be any function in V_g. (If V_g were empty, there would be no candidate u at all.) For part (d), where $g = 1$, the constant $G \equiv 1$ works, since $\iint_\Omega [1^2 + 0]\,\mathrm{d}x\,\mathrm{d}y = 4 < \infty$. Write $u = w + G$. Then $w = g - g = 0$ on $\partial\Omega$, so $w \in V_0$. Substituting into (4a.5) and moving the known part to the right-hand side gives

$$\underbrace{\iint_\Omega \nabla w \cdot \nabla v \,\mathrm{d}x\,\mathrm{d}y}_{B(w,v)} = \underbrace{\iint_\Omega f v \,\mathrm{d}x\,\mathrm{d}y - \iint_\Omega \nabla G \cdot \nabla v \,\mathrm{d}x\,\mathrm{d}y}_{\widetilde{L}(v)} \qquad \forall\, v \in V := V_0 = H_0^1(\Omega), \tag{4b.1}$$

where V carries the norm of Lecture 13, $\|v\|_V^2 = \|v\|^2 + \|\nabla v\|^2$, and $\|\cdot\|$ denotes the $L^2(\Omega)$ norm.

Symmetry (S) and continuity (CB) [Lecture 13, Part III, checks 1–2; Lecture 6: Cauchy]

(S) Since $\nabla w \cdot \nabla v = \nabla v \cdot \nabla w$ at every point, $B(w, v) = B(v, w)$.

(CB) The pointwise bound $|\boldsymbol{a} \cdot \boldsymbol{b}| \le |\boldsymbol{a}||\boldsymbol{b}|$, the Cauchy inequality, and $\|\nabla v\|^2 \le \|v\|^2 + \|\nabla v\|^2$ give in turn

$$|B(w,v)| \le \underbrace{\iint_\Omega |\nabla w|\,|\nabla v| \,\mathrm{d}x\,\mathrm{d}y}_{|\boldsymbol{a}\cdot\boldsymbol{b}| \le |\boldsymbol{a}||\boldsymbol{b}|} \le \underbrace{\|\nabla w\|\,\|\nabla v\|}_{\text{Cauchy}} \le \underbrace{\|w\|_V \|v\|_V}_{\|\nabla v\|^2 \le \|v\|^2 + \|\nabla v\|^2}, \qquad \beta^* = 1. \tag{4b.2}$$

Poincaré's inequality on the square [Lecture 8, eqs. (1.127)–(1.128): the same three lines, in x, for each fixed y]

Let $v \in V_0$ and fix y. Since $v(-1, y) = 0$ on the left side of the square,

$$v(x,y) = \int_{-1}^{x} v_{,x}(s,y)\,\mathrm{d}s. \tag{4b.3}$$

Extending the integral to the whole line and applying the Cauchy inequality to the product $1 \cdot |v_{,x}|$ on an interval of length 2 gives

$$|v(x,y)| \le \int_{-1}^{1} |v_{,x}(s,y)|\,\mathrm{d}s \le \underbrace{\Big(\int_{-1}^{1} 1^2\,\mathrm{d}s\Big)^{1/2}}_{\sqrt{2}} \Big(\int_{-1}^{1} v_{,x}(s,y)^2\,\mathrm{d}s\Big)^{1/2}. \tag{4b.4}$$

Now square (4b.4). Its right-hand side no longer depends on x, so integrating over $x \in (-1, 1)$ multiplies it by 2. Integrating then over $y \in (-1, 1)$ gives

$$\boxed{\|v\|^2 = \iint_\Omega v^2 \,\mathrm{d}x\,\mathrm{d}y \le \underbrace{2 \cdot 2}_{x\text{-length} \times (4b.4)} \iint_\Omega (v_{,x})^2 \,\mathrm{d}x\,\mathrm{d}y \le 4\,\|\nabla v\|^2} \tag{4b.5}$$

This is Lecture 13's multi-dimensional Poincaré inequality, $\|v\| \le C\|\nabla v\|$, with $C = 2$.

Ellipticity (VE) and boundedness of $\widetilde{L}$ [Lecture 13, Part III, checks 3–4]

(VE) Split $B(v,v)$ into two halves and apply (4b.5) to one of them:

$$B(v,v) = \|\nabla v\|^2 = \tfrac{1}{2}\|\nabla v\|^2 + \underbrace{\tfrac{1}{2}\|\nabla v\|^2}_{\ge \frac{1}{8}\|v\|^2 \text{ by (4b.5)}} \ge \tfrac{1}{8}\big(\|\nabla v\|^2 + \|v\|^2\big) = \tfrac{1}{8}\|v\|_V^2, \qquad \beta_* = \tfrac{1}{8}. \quad \text{(4b.6)}$$

This agrees with Lecture 13's $\beta_* = \frac{1}{2}\min(1/C^2, 1)$ for $C = 2$.

$\widetilde{L}$ bounded Applying the Cauchy inequality to each term of (4b.1) gives

$$|\widetilde{L}(v)| \le \|f\|\,\|v\| + \|\nabla G\|\,\|\nabla v\| \le \big(\|f\| + \|\nabla G\|\big)\|v\|_V. \quad \text{(4b.7)}$$

This requires $\iint_\Omega f^2 < \infty$. In part (d), f is continuous on the closed square. Since $e^{\sin \pi x \sin \pi y} \le e$ and the expression in braces has absolute value at most $2 + 2 + 1 + 1 = 6$, we have $|f| \le 6\pi^2 e$. Moreover $\nabla G = 0$.

condition	holds because	constant	equation
(S) symmetric	$\nabla w \cdot \nabla v = \nabla v \cdot \nabla w$	—	Step 2
(CB) bounded	Cauchy	$\beta^* = 1$	(4b.2)
(VE) V-elliptic	Poincaré, $C = 2$	$\beta_* = 1/8$	(4b.6)
$\widetilde{L}$ bounded	Cauchy	$\|f\| + \|\nabla G\|$	(4b.7)

Table 4: The four Lax–Milgram conditions for (4b.1) on $V = H_0^1\big((-1,1)^2\big)$.

Conclusion, and uniqueness of u itself

Table 4 shows that every hypothesis of the Lax–Milgram theorem is satisfied. Hence (4b.1) has a unique solution $w \in V_0$, with $\|w\|_V \le \big(\|f\| + \|\nabla G\|\big)/\beta_* = 8\big(\|f\| + \|\nabla G\|\big)$, and $u = w + G \in V_g$ solves the variational problem of 4(a).

The solution u does not depend on the choice of G. Suppose $u_1, u_2 \in V_g$ both solve the problem. Their difference $d = u_1 - u_2$ has boundary value $g - g = 0$, so $d \in V_0$, and subtracting the two variational equations gives $B(d, v) = 0$ for all $v \in V_0$. Choosing $v = d$ and using (4b.6),

$$\tfrac{1}{8}\|d\|_V^2 \le B(d,d) = 0 \qquad \Longrightarrow \qquad d = 0, \quad u_1 = u_2. \quad \text{(4b.8)}$$

Conclusion. With $u = w + G$, the problem for w on $H_0^1(\Omega)$ satisfies (S), (CB) with $\beta^* = 1$, (VE) with $\beta_* = 1/8$, and $\widetilde{L}$ is bounded. By Lax–Milgram the variational problem of 4(a) has exactly one solution $u \in V_g$.

Mechanics reading. (VE) is "no rigid-body mode": the rim is held, so a membrane cannot move without stretching. (4b.5) is the 2D version of Lecture 8's "pinned ends cannot move for free" — held on the left side, a deflection is controlled by its slope. $\beta^*/\beta_* = 8$ is the constant that Céa's lemma will carry into the error estimate of part (d).

Problem 4(c)

Develop a finite element method for this problem based on square elements. On each element the solution on the four corners is interpolated with $s(x,y) = \alpha_1 + \alpha_2 x + \alpha_3 y + \alpha_4 xy$. Write out the basis functions and explicitly write out the linear system (uniform grid, same spacing in x and y).

Grid and elements

The square is divided into $N \times N$ elements of side $h = 2/N$, with nodes $x_i = -1+ih$ and $y_j = -1+jh$, $i, j = 0, \ldots, N$. The $(N-1)^2$ interior nodes, $1 \le i, j \le N-1$, carry the unknowns, and the boundary nodes carry the known values $g(x_i, y_j)$. The corners of the element $\Omega_{ij} = [x_i, x_{i+1}] \times [y_j, y_{j+1}]$ are numbered counter-clockwise starting from (x_i, y_j) (Figure 3).

The basis functions on one element [the same steps as the triangle, Problem 3, eq. (3.4)]

On Ω_{ij}, let φ_p be the hat function of corner p ($p = 1, 2, 3, 4$): it equals 1 at corner p and 0 at the other three corners. It has the form of the given interpolant, $\varphi_p(x,y) = \alpha_1 + \alpha_2 x + \alpha_3 y + \alpha_4 xy$, with four coefficients, one per corner, whose values depend on p. Evaluating it at the four corners gives four equations for its coefficients:

$$\underbrace{\begin{bmatrix} 1 & x_i & y_j & x_i y_j \\ 1 & x_{i+1} & y_j & x_{i+1} y_j \\ 1 & x_{i+1} & y_{j+1} & x_{i+1} y_{j+1} \\ 1 & x_i & y_{j+1} & x_i y_{j+1} \end{bmatrix}}_{\boldsymbol{X}} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} = \begin{bmatrix} \delta_{p1} \\ \delta_{p2} \\ \delta_{p3} \\ \delta_{p4} \end{bmatrix}. \tag{4c.1}$$

This is the triangle's system (3.4) with one more corner and one more column, xy. Take $p = 1$. Subtracting rows in pairs and using $x_{i+1} - x_i = y_{j+1} - y_j = h$ gives

$$\underbrace{h\,(\alpha_2 + \alpha_4 y_j) = -1}_{\text{row 2} - \text{row 1}}, \qquad \underbrace{h\,(\alpha_2 + \alpha_4 y_{j+1}) = 0}_{\text{row 3} - \text{row 4}}, \qquad \underbrace{h\,(\alpha_3 + \alpha_4 x_i) = -1}_{\text{row 4} - \text{row 1}}. \tag{4c.2}$$

The second equation minus the first gives $h^2 \alpha_4 = 1$. Then the second gives α_2, the third gives α_3, and row 3 of (4c.1) gives α_1:

$$\alpha_4 = \frac{1}{h^2}, \qquad \alpha_2 = -\frac{y_{j+1}}{h^2}, \qquad \alpha_3 = -\frac{1}{h} - \frac{x_i}{h^2} = -\frac{x_{i+1}}{h^2},$$
$$\alpha_1 = -\alpha_2 x_{i+1} - \alpha_3 y_{j+1} - \alpha_4 x_{i+1} y_{j+1} = \frac{x_{i+1} y_{j+1}}{h^2}.$$

Hence $\varphi_1 = \big(x_{i+1}y_{j+1} - y_{j+1}x - x_{i+1}y + xy\big)/h^2 = (x_{i+1} - x)(y_{j+1} - y)/h^2$. Moving the 1 to corner $p = 2, 3, 4$ and repeating the same subtractions gives the other three hat functions:

$$\boxed{\begin{aligned} \varphi_1 &= \frac{(x_{i+1} - x)(y_{j+1} - y)}{h^2}, & \varphi_2 &= \frac{(x - x_i)(y_{j+1} - y)}{h^2}, \\ \varphi_3 &= \frac{(x - x_i)(y - y_j)}{h^2}, & \varphi_4 &= \frac{(x_{i+1} - x)(y - y_j)}{h^2} \end{aligned}} \tag{4c.3}$$

Each one has the form $\alpha_1 + \alpha_2 x + \alpha_3 y + \alpha_4 xy$, equals 1 at its own corner, and has a factor that vanishes at each of the other three corners, so it satisfies (4c.1). The subtractions in (4c.2) solve (4c.1) for any right-hand side, so $\boldsymbol{X}$ is invertible and the four corner values $s_1, \ldots, s_4$ determine the interpolant uniquely: $s = \sum_{p=1}^{4} s_p \varphi_p$.

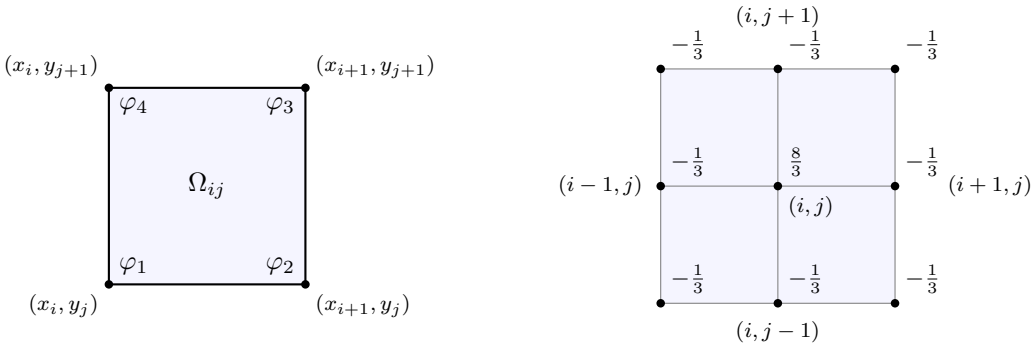


Figure 3: Left: one element Ω_{ij} and the hat functions of its corners, (4c.3). Right: the four elements around an interior node (i, j) — the support of φ_{ij} — with the coefficients of its row of the stiffness matrix, (4c.10).

Continuity and the global basis [Lecture 3, eq. (1.41): 1D hats; HW3 Problem 3: a jump puts a delta in the derivative]

On the right edge of Ω_{ij}, where $x = x_{i+1}$, (4c.3) gives $\varphi_1 = \varphi_4 = 0$, so $s = s_2(y_{j+1} - y)/h + s_3(y - y_j)/h$, which is linear in y and involves only the two corner values on that edge. The neighbouring element $\Omega_{i+1,j}$ produces the same linear function on its left edge from the same two values. Hence s is continuous across every edge, ∇s contains no delta function, and $\iint_\Omega |\nabla s|^2 < \infty$. The method is therefore **conforming**: V_h is contained in the spaces of (4a.6).

The global basis function of node (i, j) is obtained by joining the hat functions of the four squares around it (Figure 3, right). The node is corner 1 of Ω_{ij}, corner 2 of $\Omega_{i-1,j}$, corner 3 of $\Omega_{i-1,j-1}$ and corner 4 of $\Omega_{i,j-1}$. On Ω_{ij}, (4c.3) gives $(x_{i+1} - x)(y_{j+1} - y)/h^2$; on $\Omega_{i-1,j}$ it gives $(x - x_{i-1})(y_{j+1} - y)/h^2$; and similarly on the other two. In every case the x-factor, divided by h, is one half of Lecture 3's hat function φ_i on the x-grid, and the y-factor is one half of φ_j on the y-grid:

$$\varphi_{ij}(x, y) = \varphi_i(x)\,\varphi_j(y), \qquad \varphi_i(x) = \begin{cases} (x - x_{i-1})/h, & x_{i-1} \le x \le x_i, \\ (x_{i+1} - x)/h, & x_i \le x \le x_{i+1}, \\ 0, & \text{otherwise.} \end{cases} \tag{4c.4}$$

The trial function carries the boundary values as Lecture 5's lift, and the test functions are the interior basis functions:

$$U = \underbrace{\sum_{(k,l)\in\partial\Omega} g(x_k, y_l)\,\varphi_{kl}}_{\text{known, carries } u=g} + \sum_{i,j=1}^{N-1} \xi_{ij}\,\varphi_{ij}, \qquad v = \varphi_{ij}, \quad 1 \le i, j \le N-1. \tag{4c.5}$$

(FEM) Find U of the form (4c.5) such that $\displaystyle\iint_\Omega \nabla U \cdot \nabla\varphi_{ij}\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f\varphi_{ij}\,\mathrm{d}x\,\mathrm{d}y$ for every interior node (i, j).

The element stiffness matrix

Differentiating (4c.3) gives the gradients on Ω_{ij}:

$$\begin{aligned} \nabla\varphi_1 &= \tfrac{1}{h^2}\big(-(y_{j+1} - y),\, -(x_{i+1} - x)\big), & \nabla\varphi_2 &= \tfrac{1}{h^2}\big(y_{j+1} - y,\, -(x - x_i)\big), \\ \nabla\varphi_3 &= \tfrac{1}{h^2}\big(y - y_j,\, x - x_i\big), & \nabla\varphi_4 &= \tfrac{1}{h^2}\big(-(y - y_j),\, x_{i+1} - x\big). \end{aligned} \tag{4c.6}$$

Unlike on the triangle, these are not constant, so the step "a constant leaves the integral" of (3.5) is replaced by integration over Ω_{ij}. Each component in (4c.6) depends on only one of x and y, so every integral separates into one-dimensional integrals over an interval of length h:

$$\int_{x_i}^{x_{i+1}} \mathrm{d}x = h, \qquad \int_{x_i}^{x_{i+1}} (x_{i+1}-x)^2\,\mathrm{d}x = \int_{x_i}^{x_{i+1}} (x-x_i)^2\,\mathrm{d}x = \frac{h^3}{3},$$
$$\int_{x_i}^{x_{i+1}} \underbrace{(x_{i+1}-x)(x-x_i)}_{h(x-x_i)-(x-x_i)^2}\,\mathrm{d}x = \frac{h^3}{2} - \frac{h^3}{3} = \frac{h^3}{6}, \tag{4c.7}$$

and the same in y. The entries of the element stiffness matrix are $k^e_{pq} = \iint_{\Omega_{ij}} \nabla\varphi_p \cdot \nabla\varphi_q \,\mathrm{d}x\,\mathrm{d}y$, and there are three kinds: a corner with itself, two corners that share an edge, and two opposite corners. Taking the dot products in (4c.6) and integrating with (4c.7),

$$\begin{aligned}
k^e_{11} &= \frac{1}{h^4}\iint_{\Omega_{ij}} \big[(y_{j+1}-y)^2 + (x_{i+1}-x)^2\big]\,\mathrm{d}x\,\mathrm{d}y = \frac{1}{h^4}\Big[h\cdot\frac{h^3}{3} + \frac{h^3}{3}\cdot h\Big] = \frac{4}{6},\\
k^e_{12} &= \frac{1}{h^4}\iint_{\Omega_{ij}} \big[-(y_{j+1}-y)^2 + (x_{i+1}-x)(x-x_i)\big]\,\mathrm{d}x\,\mathrm{d}y = \frac{1}{h^4}\Big[-h\cdot\frac{h^3}{3} + \frac{h^3}{6}\cdot h\Big] = -\frac{1}{6},\\
k^e_{13} &= \frac{1}{h^4}\iint_{\Omega_{ij}} \big[-(y_{j+1}-y)(y-y_j) - (x_{i+1}-x)(x-x_i)\big]\,\mathrm{d}x\,\mathrm{d}y = \frac{1}{h^4}\Big[-h\cdot\frac{h^3}{6} - \frac{h^3}{6}\cdot h\Big] = -\frac{2}{6}.
\end{aligned} \tag{4c.8}$$

The factor h^4 cancels: in two dimensions the stiffness of a square element does not depend on its size. The matrix is symmetric, $k^e_{pq} = k^e_{qp}$, and a quarter turn of the square maps corner $1 \to 2 \to 3 \to 4 \to 1$, so every other entry repeats one of these three values, and

$$\boxed{\boldsymbol{k}^e = \frac{1}{6}\begin{bmatrix} 4 & -1 & -2 & -1 \\ -1 & 4 & -1 & -2 \\ -2 & -1 & 4 & -1 \\ -1 & -2 & -1 & 4 \end{bmatrix}} \tag{4c.9}$$

Each row sums to 0, because a constant function has no gradient and therefore no energy.

The linear system [Lecture 13, eq. (1.236): $A_{ij} = B(\varphi_i, \varphi_j)$; Lecture 5, eq. (1.66): known values to the RHS; Lecture 4: trapezoidal load]

Row of an interior node. An interior node (i, j) is a corner of 4 squares, shares each of its edge-neighbours with 2 squares, and shares each of its diagonal neighbours with 1 square. By (4c.8), its diagonal entry is $4 \cdot \frac{4}{6} = \frac{8}{3}$, an edge-neighbour entry is $2 \cdot (-\frac{1}{6}) = -\frac{1}{3}$, and a diagonal-neighbour entry is $1 \cdot (-\frac{2}{6}) = -\frac{1}{3}$. The equation of node (i, j) is therefore

$$\boxed{\frac{1}{3}\Big[\,8\,U_{i,j} - \sum_{(k,l)\in\mathcal{N}_{ij}} U_{k,l}\Big] = b_{ij}, \qquad 1 \le i, j \le N-1,} \tag{4c.10}$$

where $\mathcal{N}_{ij}$ is the set of eight neighbours (k, l) with $k \in \{i-1, i, i+1\}$, $l \in \{j-1, j, j+1\}$ and $(k, l) \neq (i, j)$, and $U_{k,l} = g(x_k, y_l)$ whenever (k, l) is a boundary node.

Load vector. The load entry is $b_{ij} = \iint_\Omega f\varphi_{ij}$. Lecture 4's trapezoidal rule on each square, $\iint_{\Omega_{ij}} F \approx \frac{h^2}{4}\sum_{\text{corners}} F$, sees $f\varphi_{ij}$ only at the corner (i,j), where $\varphi_{ij} = 1$. Summing over the four squares gives

$$b_{ij} \approx 4 \cdot \tfrac{h^2}{4}\, f(x_i, y_j) = h^2 f_{ij}. \tag{4c.11}$$

Matrix form. Order the unknowns row by row, $\boldsymbol{U} = (U_{1,1}, \dots, U_{N-1,1}, U_{1,2}, \dots, U_{N-1,N-1})^T$, and move every boundary value in (4c.10) to the right-hand side. The system becomes

$$\boxed{\begin{gathered} \boldsymbol{A}\,\boldsymbol{U} = \boldsymbol{b} + \boldsymbol{r}, \qquad \boldsymbol{A} = \frac{1}{3}\begin{bmatrix} \boldsymbol{T} & \boldsymbol{S} & & \\ \boldsymbol{S} & \boldsymbol{T} & \ddots & \\ & \ddots & \ddots & \boldsymbol{S} \\ & & \boldsymbol{S} & \boldsymbol{T} \end{bmatrix}, \\ \boldsymbol{T} = \begin{bmatrix} 8 & -1 & & \\ -1 & 8 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 8 \end{bmatrix}, \qquad \boldsymbol{S} = \begin{bmatrix} -1 & -1 & & \\ -1 & -1 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & -1 \end{bmatrix} \end{gathered}} \tag{4c.12}$$

Here $\boldsymbol{A}$ is $(N-1)^2 \times (N-1)^2$ and consists of $N-1$ block rows. The blocks $\boldsymbol{T}$ and $\boldsymbol{S}$ are $(N-1)\times(N-1)$: $\boldsymbol{T}$ couples a node to its own grid row, and $\boldsymbol{S}$ couples it to the rows above and below. The boundary vector has entries $r_{ij} = \frac{1}{3}\sum g(x_k, y_l)$, summed over the boundary nodes among the eight neighbours of (i,j). For $g = 1$, as in part (d),

$$r_{ij} = \begin{cases} 5/3, & (i,j) \text{ next to a corner (5 boundary neighbours)}, \\ 1, & (i,j) \text{ next to one side (3 boundary neighbours)}, \\ 0, & \text{otherwise.} \end{cases} \tag{4c.13}$$

Finally, $\boldsymbol{A}$ is symmetric positive definite by Lecture 13's argument, since (S) and (VE) of 4(b) hold whatever the element shape.

Same as finite differences? Divide (4c.10) by h^2 and use (4c.11): $\big[8U_{i,j} - \sum_{\mathcal{N}_{ij}} U_{k,l}\big]/(3h^2) = f_{ij}$, a nine-point difference formula for $-\Delta u$. This is Lecture 4's "CG(1) with trapezoidal load is the central difference method", one dimension up: the 2D element couples diagonal neighbours too, which the five-point formula does not.

Problem 4(d)

Use the method of 4(c) to solve $-\nabla\cdot\nabla u = f$ *on* $\Omega=[-1,1]^2$, $u=g$ *on* $\partial\Omega$, *with*

$$f=\pi^2 e^{\sin(\pi x)\sin(\pi y)}\big\{2\sin(\pi x)\sin(\pi y)+2\cos^2(\pi x)\cos^2(\pi y)-\cos^2(\pi x)-\cos^2(\pi y)\big\},\qquad g=1.$$

Create a convergence table showing both $\|\cdot\|_V$ *and* $\|\cdot\|_{L^2(\Omega)}$ *errors. What order of convergence do you observe? The exact solution is* $u=e^{\sin(\pi x)\sin(\pi y)}$.

Consistency of the data

With $s=\sin\pi x\sin\pi y$ and $u=e^s$, we have $u_{,x}=\pi\cos\pi x\sin\pi y\,e^s$, and differentiating once more in x and in y gives

$$\begin{aligned}-u_{,xx}-u_{,yy}&=\pi^2e^s\big[2\sin\pi x\sin\pi y-\cos^2\pi x\ \underbrace{\sin^2\pi y}_{1-\cos^2\pi y}\ -\ \underbrace{\sin^2\pi x}_{1-\cos^2\pi x}\cos^2\pi y\big]\\&=\pi^2e^s\big[2\sin\pi x\sin\pi y+2\cos^2\pi x\cos^2\pi y-\cos^2\pi x-\cos^2\pi y\big]=f.\end{aligned}\tag{4d.1}$$

This identity was also checked symbolically ($-\Delta u-f\equiv0$). On every side of the square $\sin(\pm\pi)=0$, so $s=0$ and $u=e^0=1=g$.

Implementation [4(c); Lecture 5, eq. (1.66)]

The grids have $N\times N$ squares with $N=4,\dots,256$, so $h=2/N$ and there are $(N-1)^2$ unknowns. The stiffness matrix is assembled from the element matrix $\boldsymbol{k}^e$ of 4(c), which is the same on every square. The code checks that every interior row is the stencil $\frac{1}{3}[8;-1\times8]$ of 4(c), that the interior matrix is symmetric positive definite, and that the boundary terms are 5/3, 1 and 0 as in 4(c). The boundary values $U=g=1$ are moved to the right-hand side, as in Lecture 5.
The load $b_{ij}=\iint f\varphi_{ij}$ is computed with 3×3 Gauss points per square rather than with the trapezoidal value h^2f_{ij} of 4(c); the two are compared below. The errors are computed with 4×4 Gauss points per square, in the norms of Lecture 13, eq. (1.254):

$$\|e\|^2_{L^2(\Omega)}=\iint_\Omega(u-U)^2\,\mathrm{d}x\,\mathrm{d}y,\qquad \|e\|^2_V=\|e\|^2_{L^2(\Omega)}+\iint_\Omega|\nabla(u-U)|^2\,\mathrm{d}x\,\mathrm{d}y.\tag{4d.2}$$

The observed rate is $\log_2\big(E(h)/E(h/2)\big)$, where E is either error.

Convergence table

N	h	unknowns	$\|e\|_V$	rate	$\|e\|_{L^2(\Omega)}$	rate
4	0.50000	9	2.1378×10^{0}	—	2.3519×10^{-1}	—
8	0.25000	49	1.4636×10^{0}	0.55	9.5737×10^{-2}	1.30
16	0.12500	225	7.3914×10^{-1}	0.99	2.4256×10^{-2}	1.98
32	0.06250	961	3.7048×10^{-1}	1.00	6.0808×10^{-3}	2.00
64	0.03125	3969	1.8536×10^{-1}	1.00	1.5213×10^{-3}	2.00
128	0.01562	16129	9.2693×10^{-2}	1.00	3.8038×10^{-4}	2.00
256	0.00781	65025	4.6348×10^{-2}	1.00	9.5100×10^{-5}	2.00

Table 5: Bilinear squares on $[-1,1]^2$, $N\times N$ elements.

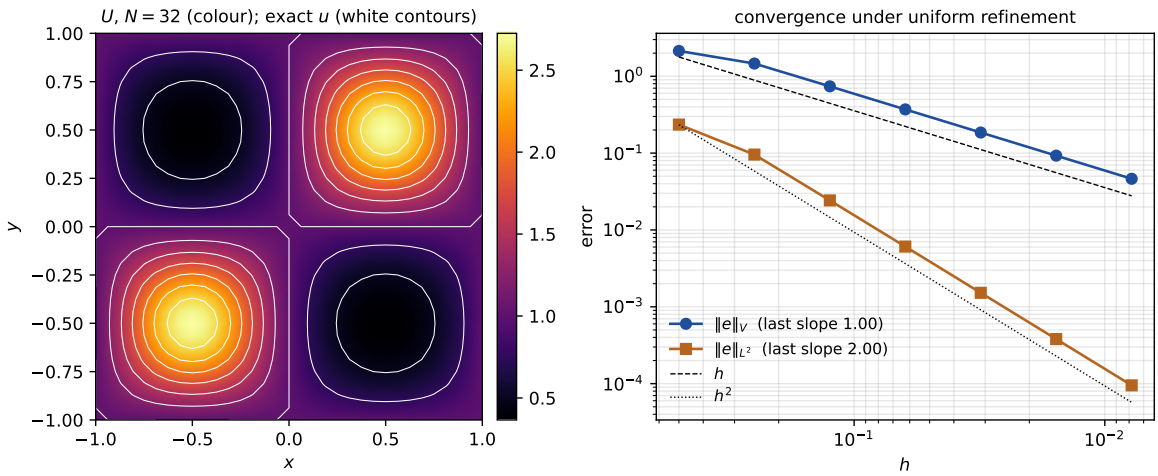


Figure 4: Left: U at $N = 32$ with contours of the exact u. Right: the errors of Table 5 against h, with slopes 1 and 2 for reference.

The observed orders

Table 5 shows the rate in $\|e\|_V$ tending to 1.00, that is, $\|e\|_V = \mathcal{O}(h)$, and the rate in $\|e\|_{L^2(\Omega)}$ tending to 2.00, that is, $\|e\|_{L^2(\Omega)} = \mathcal{O}(h^2)$. The rates are clean from $N = 16$ on; at $N = 4$ and 8 the mesh has only two to four squares across each bump of u. The trapezoidal load of 4(c) gives the same rates. At $N = 256$ it yields $\|e\|_V = 4.6355 \times 10^{-2}$ and $\|e\|_{L^2} = 9.7057 \times 10^{-5}$, against 4.6348×10^{-2} and 9.5100×10^{-5} with Gauss quadrature.

The first-order rate in $\|\cdot\|_V$ follows from Céa's lemma. The boundary terms of 4(c) reproduce $g = 1$ exactly, so $U - 1$ is the Galerkin approximation of $w = u - 1 \in H^1_0$, and Céa's lemma (Lecture 13, eq. (1.242)) with the constants of 4(b) gives

$$\|u - U\|_V \le \frac{\beta^*}{\beta_*} \|u - v\|_V = 8 \|u - v\|_V \qquad \text{for every admissible } v \in V_h. \tag{4d.3}$$

Taking v to be the bilinear interpolant of u makes the right-hand side $\mathcal{O}(h)$. The second-order rate in L^2 follows from Lecture 7's duality argument, in which the interpolation estimate is used twice, once for e and once for the dual solution, contributing one factor of h each.

NOT FROM LECTURE — the interpolation estimate for squares

Lecture 13 states $\|u - u_h\|_{H^1} \le c \|h\, u\|_{H^2}$, eq. (1.262), for linear triangles. The same $\mathcal{O}(h)$ bound for the bilinear interpolant on squares, and the 2D duality argument behind $\mathcal{O}(h^2)$ in L^2, are standard but were not shown in class. Table 5 confirms both rates.

Conclusion. Bilinear squares converge at $\mathcal{O}(h)$ in $\|\cdot\|_V$ and $\mathcal{O}(h^2)$ in $\|\cdot\|_{L^2(\Omega)}$ — the same orders as CG(1) — with $\|e\|_V = 4.63 \times 10^{-2}$ and $\|e\|_{L^2} = 9.51 \times 10^{-5}$ at $N = 256$.